

Sydney Technical High School



TRIAL HIGHER SCHOOL CERTIFICATE

2007

MATHEMATICS EXTENSION 2

General Instructions

- Reading time - 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Approved calculators may be used
- All necessary working should be shown in every question
- A table of standard integrals is supplies at the back of this paper
- Start each question on a new page
- Attempt all Questions 1 – 8
- All questions are of equal value
- **Total marks 120**

Name: _____

Class: _____

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Question 7	Question 8	TOTAL

QUESTION 1 (15 Marks)**Marks**

a) Find by using a suitable substitution or otherwise

i) $\int \frac{dx}{\sqrt{9-16x^2}}$ 2

ii) $\int \frac{dx}{\sqrt{x^2+6x+13}}$ 2

iii) $\int \sec^3 x \tan x \, dx$ 2

b) Using the substitution $x = 3 \tan \theta$ or otherwise find $\int \frac{dx}{(9+x^2)^{\frac{3}{2}}}$ 4

c) i) Show that $\frac{d}{dx} \left[\frac{1}{2a} \log_e \left(\frac{x-a}{x+a} \right) \right] = \frac{1}{x^2 - a^2}$ 2

ii) Hence by using the substitution $x = u^2$ or otherwise find $\int \frac{\sqrt{x}}{x-1} \, dx$ 3

Question 2 (15 marks)

a) Find d if $(3+2i)(4-di)$ is wholly imaginary 2

b) If $\alpha = -2+2\sqrt{3}i$ and $\beta = 1-i$

i) Find $\frac{\alpha}{\beta}$ in the form $x+iy$ 1

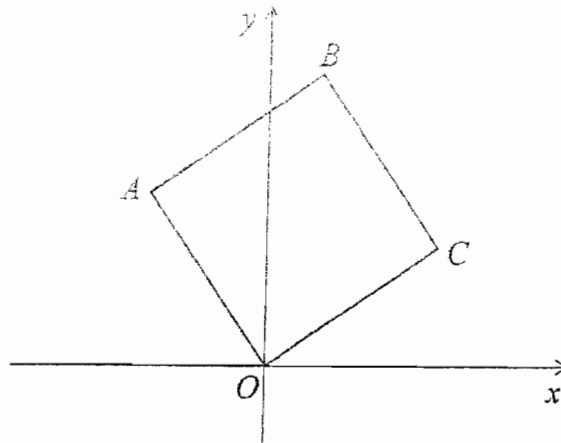
ii) Express α in modulus – argument form 1

iii) Given $\beta = \sqrt{2} \left(\cos\left(\frac{-\pi}{4}\right) + i \sin\left(\frac{-\pi}{4}\right) \right)$ find the modulus- argument form of $\frac{\alpha}{\beta}$ 2

iv) Hence find the exact value of $\cos\left(\frac{\pi}{12}\right)$ 2

c)

Marks



On the Argand diagram above, $OABC$ is a square. If B represents the complex number $4 + 6i$ find the complex number represented by C .

3

- d) i) Sketch the region in the complex number plane where the inequalities 2

$$|z - 1| \leq |z - i| \text{ and } |z - 2 - 2i| \leq 1 \text{ hold simultaneously}$$

- ii) If P is a point on the boundary of this region representing the complex number

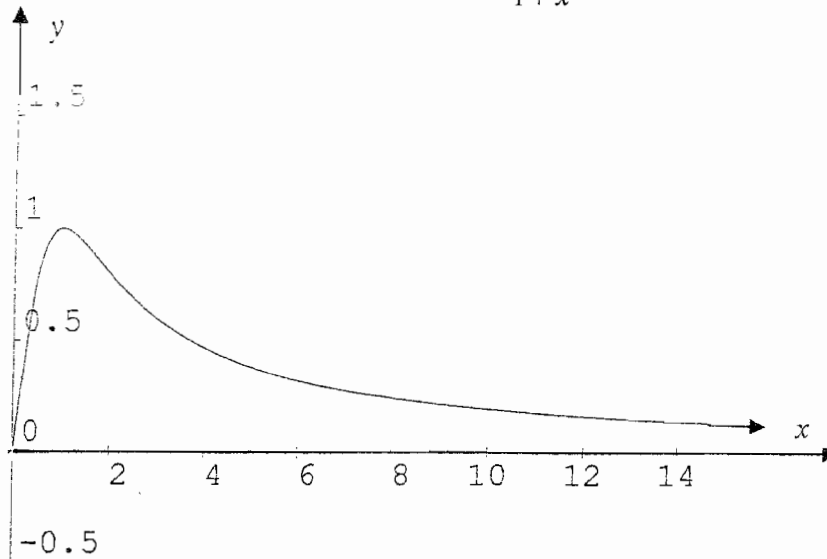
$$z, \text{ find the values of } z \text{ in the form } x + iy \text{ where } \arg(z - 1) = \frac{\pi}{4}$$

2

Question 3 (15 marks)

Marks

- a) The diagram shows the graph of $f(x) = \frac{2x}{1+x^2}$ for $x \geq 0$



For each of the following draw a one-third page sketch:

- | | | |
|------|--|---|
| i) | Sketch the graph of $y = \frac{2x}{1+x^2}$ for all real x | 1 |
| ii) | Use your completed graph in (i) to help sketch the graphs of | |
| α) | $y = \frac{ 2x }{1+x^2}$ | 2 |
| β) | $y^2 = \frac{2x}{1+x^2}$ | 2 |
| γ) | $y = \log_e \left[\frac{2x}{1+x^2} \right]$ | 2 |
| iii) | Sketch $y = \frac{1+x^2}{2x}$ clearly showing and stating the equations of any asymptotes. | 2 |
| iv) | Find the value(s) of A so that the graphs of $y = \frac{Ax}{1+x^2}$ and $y = \frac{1+x^2}{Ax}$ have no points in common. | 2 |
- b) The area between the curve $y = \frac{2x}{1+x^2}$ and the x -axis for $0 \leq x \leq 1$ is rotated about the y -axis. Use the method of cylindrical shells to find the volume of the resulting solid of revolution

Question 4 (15 marks)

Marks

- a) $P(a \cos \theta, b \sin \theta)$ and $Q(a \cos(-\theta), b \sin(-\theta))$ are the extremities of the latus rectum, $x = ae$, of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

i) Draw a neat diagram, marking the points P and Q and clearly showing the angle θ .

1

ii) Show that $\cos \theta = e$

1

iii) Show that the length of PQ is $\frac{2b^2}{a}$

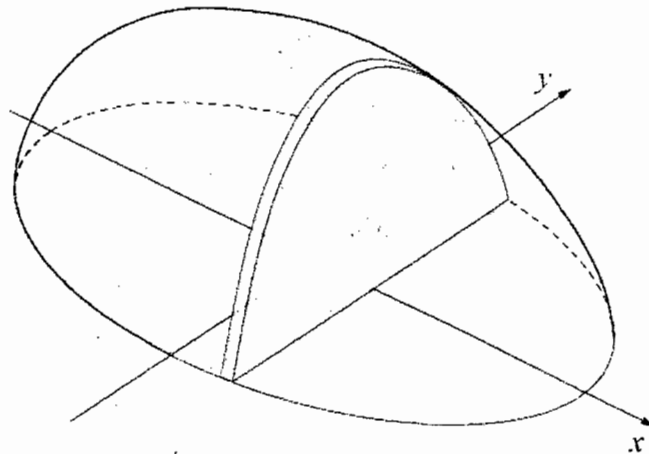
2

- b) Show that the area enclosed between the parabola $x^2 = 4ay$ and its latus rectum is $\frac{8a^2}{3}$ units²

3

- c) A solid figure has as its base, in the xy plane, the ellipse $\frac{x^2}{16} + \frac{y^2}{4} = 1$.

Cross-sections perpendicular to the x -axis are parabolas with latus rectums in the xy plane



- i) Show that the area of the cross-section at $x = h$ is $\frac{16-h^2}{6}$ units².

3

[use your answer to part (b)]

- ii) Hence, find the volume of this solid.

2

- d) Over the complex field $P(x) = 2x^3 - 15x^2 + Cx - D$ has a zero $x = 3 - 2i$

i) Determine the other two zeros

2

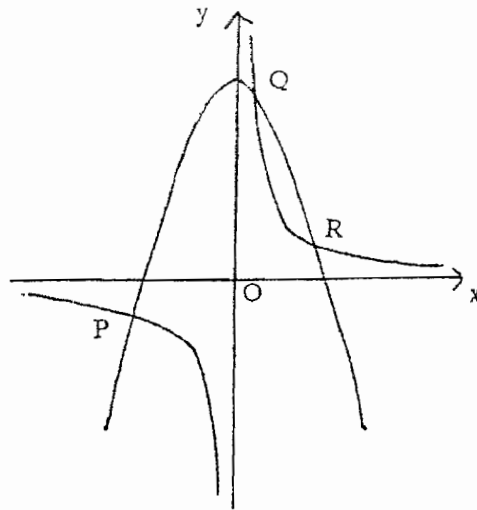
ii) Find the value of D

1

Question 5 (15 marks)

- a) The roots of the equation $z^5 - 1 = 0$ are $1, w, w^2, w^3, w^4$
- i) Mark this information on an Argand diagram 1
- ii) Find a real quadratic equation with roots $w + w^4$ and $w^2 + w^3$ 2
- iii) Hence find the value of $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5}$ 2

b)

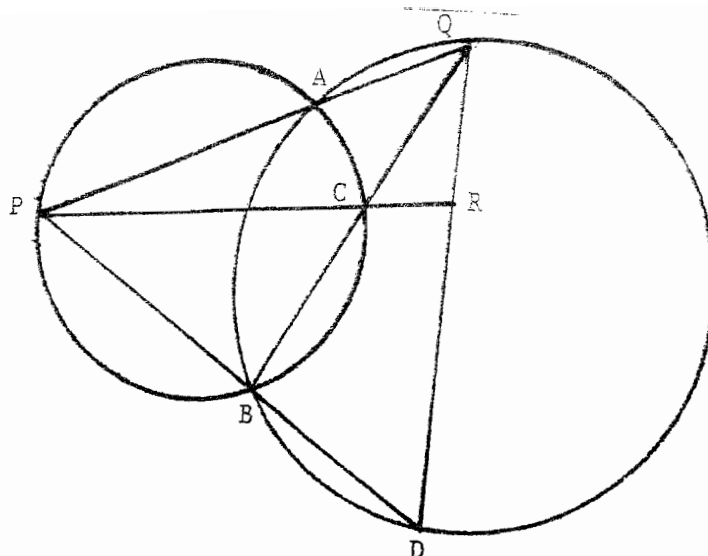


The curves $y = k - x^2$, for some real number k , and $y = \frac{1}{x}$ intersect at the points P, Q and R where $x = \alpha$, $x = \beta$ and $x = \gamma$.

- i) Show that the monic cubic equation with coefficients in terms of k whose roots are α^2 , β^2 and γ^2 is given by $x^3 - 2kx^2 + k^2x - 1 = 0$ 3
- ii) Find the monic cubic equation with coefficients in terms of k whose roots are $\frac{1}{\alpha^2}$, $\frac{1}{\beta^2}$ and $\frac{1}{\gamma^2}$ 2
- iii) Hence show that $OP^2 + OQ^2 + OR^2 = k^2 + 2k$, where O is the origin 2

c)

Marks

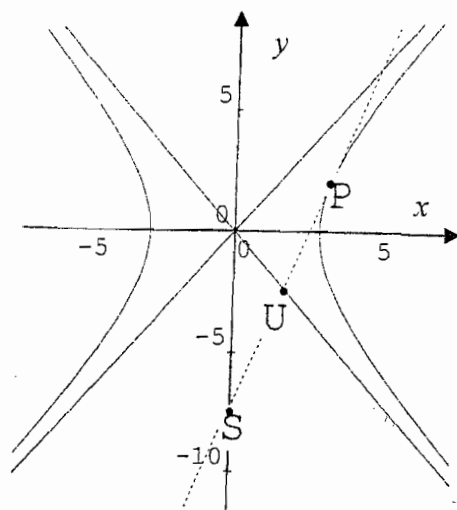


- Copy the diagram onto your page.
- Prove $BCRD$ is a cyclic quadrilateral (Hint: let $\angle D = \theta$)

3

Question 6 (15 marks)

a)



Consider the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$

- Write down the equation of each asymptote 1
- By differentiation find the gradient of the tangent to the hyperbola at $P(3 \sec \theta, 4 \tan \theta)$ 1
- Show that the equation of the tangent at P is $4x = 3 \sin \theta y + 12 \cos \theta$ 2
- Find the x -coordinate of U , the point where the tangent meets the asymptote (as shown on the diagram). 2
- Using the x -values only, find the value for θ such that U is the mid point of PS . 2

- b) i) Show that $\int_0^{\frac{\pi}{4}} \tan \theta \, d\theta = \frac{1}{2} \log_e 2$ 2
- ii) If $I_n = \int_0^{\frac{\pi}{4}} \tan^n \theta \, d\theta$ show that for $n \geq 2$ 3
- $$I_n + I_{n-2} = \frac{1}{n-1}$$
- iii) Hence evaluate $\int_0^{\frac{\pi}{4}} \tan^5 \theta \, d\theta$ 2

Question 7 (15 marks)

- a) i) Show that $\tan^{-1}(3) - \tan^{-1}(\frac{1}{2}) = \frac{\pi}{4}$ 2
- ii) Prove by mathematical induction that
- $$\sum_{r=1}^n \tan^{-1}\left(\frac{1}{2r^2}\right) = \tan^{-1}(2n+1) - \frac{\pi}{4}$$
- is true for all integral values of n for $n \geq 1$ 4
- b) A particle is moving in a straight line. After time t seconds it has displacement x metres from a fixed point O on the line, velocity $v = \frac{1-x^2}{2} \text{ ms}^{-1}$ and acceleration $a \text{ ms}^{-2}$. Initially the particle is at O .
- i) Find an expression for a in terms of x 1
- ii) Show that $\frac{2}{1-x^2} = \frac{1}{1+x} + \frac{1}{1-x}$ and hence find an expression for x in terms of t . 3
- iii) Describe the motion of the particle, explaining whether it moves to the left or right of O , whether it slows down or speeds up, and where its limiting position is. 2
- c) i) Differentiate $x^3 + y^3 = 6xy$ to find $\frac{dy}{dx}$. 1
- ii) Find the x value(s) of the point(s) where $\frac{dy}{dx} = 0$ 2

Question 8	(15 marks)	Marks
a)	i) If $S = 1 - x + x^2 - x^3 + \dots$ where $ x < 1$, find an expression for S , the limiting sum, of the series.	1
	ii) By integrating both sides of this expression and then making a substitution for x show that $\log_e 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$	2
b)	i) Show that $\int x \tan^{-1} x \, dx = \frac{1}{2}(x^2 + 1) \tan^{-1} x - \frac{1}{2}x + c$	3
	ii) If $I_n = \int_0^1 x^n \tan^{-1} x \, dx$ for $n \geq 2$ show that $I_n = \frac{\pi}{2(n+1)} - \frac{1}{n(n+1)} - \frac{n-1}{n+1} I_{n-2}$	4
c)	i) Write the general solution to $\cos 5\theta = \cos A$	1
	ii) Hence or otherwise find the total number of solutions to the equation $\cos 5\theta = \sin \theta$ for $0 \leq \theta \leq 10\pi$	4

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1

$$a) \int \frac{1}{\sqrt{9-16x^2}} dx = \frac{1}{4} \sin^{-1} \frac{4x}{3} + c$$

$$b) \int \frac{dx}{\sqrt{(x+3)^2+4}} = \ln(x+3+\sqrt{(x+3)^2+4}) + c$$

$$c) \int \sec^3 x \tan x = \int \sec^2 x \frac{d(\sec x)}{dx} dx$$

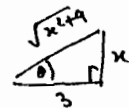
$$= \frac{1}{3} \sec^3 x + c$$

$$b) x = 3 \tan \theta, dx = 3 \sec^2 \theta d\theta$$

$$\int \frac{dx}{(9+x^2)^{3/2}} = \int \frac{3 \sec^2 \theta d\theta}{(9+9 \tan^2 \theta)^{3/2}}$$

$$\text{on simplification} = \frac{1}{9} \int \cos \theta d\theta$$

$$= \frac{1}{9} \sin \theta$$



$$= \frac{1}{9} \frac{x}{\sqrt{x^2+9}}$$

$$c) \frac{d}{dx} \left[\frac{1}{2a} \log_e \left(\frac{x-a}{x+a} \right) \right] = \frac{1}{2a} \frac{d}{dx} [\log(x-a) - \log(x+a)]$$

$$= \frac{1}{2a} \left[\frac{1}{x-a} - \frac{1}{x+a} \right]$$

$$\text{on simplification} = \frac{1}{x^2-a^2}$$

$$d) x = u^2, dx = 2u du$$

$$\int \frac{\sqrt{u} du}{x-1} = \int \frac{u \cdot 2u du}{u^2-1}$$

$$= 2 \int \frac{u^2-1+1}{u^2-1} du$$

$$= 2 \left[u + \frac{1}{2} \log_e \left(\frac{u-1}{u+1} \right) \right] + c$$

$$= 2\sqrt{x} + \log_e \left(\frac{\sqrt{x}-1}{\sqrt{x}+1} \right) + c$$

Question 2

$$a) (3+2i)(4-di) = (12+2d)$$

$$\therefore 12+2d=0$$

$$d=-6$$

$$b) i) \frac{d}{\beta} = \frac{(-1-\sqrt{3}) + i(\sqrt{3}-1)}{\beta}$$

$$ii) \alpha = 4 \cos 2\pi/3$$

$$iii) \frac{\alpha}{\beta} = \frac{4 \cos 2\pi/3}{\sqrt{2} \cos \pi/4}$$

$$= 2\sqrt{2} \cos(11\pi/12)$$

$$iv) 2\sqrt{2} \cos 11\pi/12 = -1 - \frac{1}{\sqrt{2}}$$

$$\cos 11\pi/12 = \frac{-1-\frac{1}{\sqrt{2}}}{2}$$

$$\therefore \cos \pi/12 = \frac{1+\sqrt{3}}{2\sqrt{2}}$$

$$c) C = x+iy$$

$$\therefore A = i(x+iy)$$

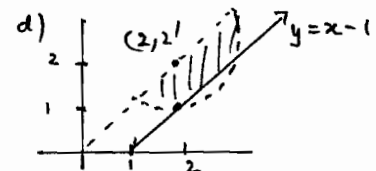
$$= ix - y$$

$$B = C+A$$

$$4+6i = x-y + i(x+y)$$

$$\therefore x=5, y=1$$

$$A = 5+i$$



$$ii) P \text{ is point of intersection of } y=x-1 \text{ and } (x-2)^2+(y-2)^2=1$$

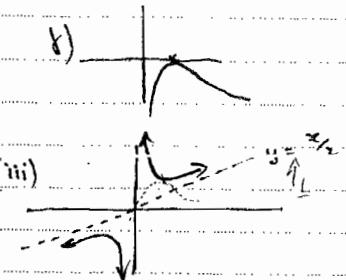
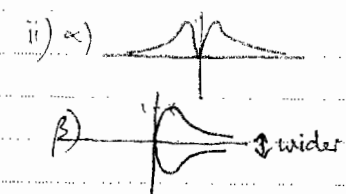
$$\therefore (x-2)^2+(x-3)^2=1$$

$$x^2-5x+6=0$$

$$(x-3)(x-2)$$

$$x=2, y=1 \quad x=3, y=2$$

$$\therefore P \text{ is } 2+i \text{ or } 3+2i$$



$$\frac{Ax}{1+x^2} = \frac{1+x^2}{Ax}$$

$$\frac{Ax}{1+x^2} = \pm 1$$

$$x^2 - Ax + 1 = 0 \text{ or } x^2 + Ax + 1 = 0$$

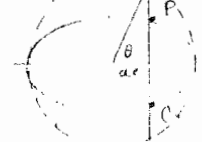
point of intersection $\Delta < 0$

solving $A^2 < 4$

$$\therefore -2 < A < 2, A \neq 0$$

$$\begin{aligned} \text{Volume} &= \lim_{\Delta x \rightarrow 0} \sum_{i=0}^1 2\pi x_i \Delta x \\ &= 4\pi \int_0^1 \frac{x^2}{1+x^2} dx \\ &= 4\pi \int_0^1 \frac{1+x^2}{1+x^2} - \frac{1}{1+x^2} dx \\ &= 4\pi \left[x - \tan^{-1} x \right]_0^1 \\ &= 4\pi \left[1 - \frac{\pi}{4} \right] \\ &= 4\pi - \pi^2 \end{aligned}$$

vi) $\frac{1}{2} \times 100 = 50\%$



$$\frac{d}{dt} \left(\frac{1}{2} m v^2 \right) = \frac{1}{2} m \frac{d}{dt} (v^2) = \frac{1}{2} m \frac{d}{dt} (v_x^2 + v_y^2 + v_z^2) = \frac{1}{2} m \frac{d}{dt} (v^2)$$

iii) $\int_{\mathbb{R}^n} \varphi(x) dx = 1$ and $\varphi(x) \geq 0$.

Fig. 2. $\frac{1}{\rho} \ln \frac{1}{1-\rho}$ vs. $\frac{1}{\rho}$ for the $\text{H}_2\text{O}-\text{H}_2\text{SO}_4$ system.

$$= 26 \int_1^{\infty} r^{-5}$$

$$b' = a' \cdot (1 - c')$$

= 24.1

243

b) i)

$$x^2 \leq 4ax$$

10

$$A: v^2 = 4a^2 - \frac{1}{2a} \frac{2^2}{3} + \dots$$

c) at $x = l_1$.

$$\frac{h^2}{4m} + \frac{h^2}{4m} = 1$$

$$y = \pm 2\sqrt{1 - \frac{1}{4}}$$

$$\therefore 2a = 2\sqrt{1 - \frac{b^2}{c^2}}$$

$$a = \sqrt{1 - \frac{v^2}{c^2}}$$

$$= 8 \cdot \left(1 - \frac{1}{2^2}\right) = 15 \cdot \frac{1}{2}$$

$$= 15 \cdot 10^3$$

$$\begin{aligned} \text{iii) Volume} &= \lim_{h \rightarrow 0} \frac{1}{3} \pi (16 - h^2) \Delta h \\ &= \frac{\pi}{3} \lim_{h \rightarrow 0} (16h - h^3) \\ &= \frac{\pi}{3} \left[16h - \frac{h^3}{3} \right]_0^4 \\ &= \frac{128\pi}{3} \end{aligned}$$

$\Delta_{11} x = 3 + 2i$ (conjugate root) theorem.
 $3 + 2i + 3 - 2i = 6$ (S.S.)

$$z - \lambda = 3/2.$$

$$\text{ii) } \sum_{i=1}^n i^2 = (3+2n)(3+2n+1) \times \frac{1}{6} = \frac{1}{6} n(2n+3)(n+2)$$

115

$$I_n = \frac{\pi}{4} - \frac{1}{2} - \frac{(n-1)}{2} [I_n + I_{n-2}] + \frac{(n-1)}{2} \int_0^1 x^{n-1} dx$$

$$I_n = \frac{\pi}{4} - \frac{1}{2} - \frac{(n-1)}{2} [I_n + I_{n-2}] + \frac{(n-1)}{2} \left[\frac{x^n}{n} \right]_0^1$$

$$2I_n = \frac{\pi}{2} - 1 - (n-1)I_n - (n-1)I_{n-2} + \frac{n-1}{n}$$

$$(n+1)I_n = \frac{\pi}{2} - \frac{1}{n} - (n-1)I_{n-2}$$

$$\therefore I_n = \frac{\pi}{2(n+1)} - \frac{1}{n(n+1)} - \frac{n-1}{n+1} I_{n-2}$$

c) i) $\cos 5\theta = \cos A$
 $5\theta = 2n\pi \pm A$

ii) $\cos 5\theta = \cos(\pi/2 - \theta)$
 $5\theta = 2n\pi \pm (\pi/2 - \theta)$

$\therefore 5\theta = 2n\pi + \pi/2 - \theta$ or $5\theta = 2n\pi - \pi/2 + \theta$
 $6\theta = \frac{4n\pi + \pi}{2}$ $4\theta = \frac{4n\pi - \pi}{2}$

$$\theta = \frac{\pi(4n+1)}{12}$$

$$\theta = \frac{\pi(4n-1)}{8}$$

Now $0 \leq \frac{\pi(4n+1)}{12} \leq 10\pi$

$0\pi \leq \frac{\pi(4n-1)}{8} \leq 10\pi$

$$0 \leq 4n+1 \leq 120$$

$$0 \leq 4n-1 \leq 80$$

$$4n \leq 119$$

$$4n \leq 81$$

$$n \leq 29^{3/4}$$

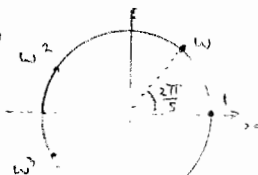
$$n \leq 20^{1/4}$$

$$\therefore n = 29$$

$$n = 20$$

$\therefore 49$ solutions (there are no common solutions)

Q. 5a)



ii) $(w+w^4) + (w^2+w^3) = \dots$
 $(w+w^4)(w^2+w^3) = w^3+w^4+w^6+w^7$
 $= w^3+w^4+w^1+w^2$
 $= -1$

$$\therefore x^2 + x - 1 = 0$$

iii) $(w+w^4) + (w^2+w^3) = 2\cos 2\pi/5 + 2\cos 4\pi/5$
 $\therefore \cos 2\pi/5 + \cos 4\pi/5 = -1/2$

b)

$$i) kx^2 = \frac{1}{x}$$

$\therefore x^3 - kx + 1 = 0$ has roots α, β, γ
 $\therefore x^3 - kx^2 + k^2x - 1 = 0$ (A)

has roots $\alpha^2, \beta^2, \gamma^2$

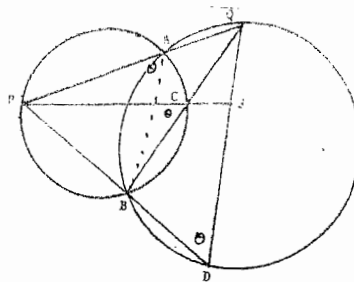
let $y = \frac{1}{x}$ in A $\Rightarrow \alpha = \frac{1}{y}$

$$\therefore x^3 - k^2x^2 + 2kx - 1 = 0$$
 (B)

has roots $\frac{1}{\alpha^2}, \frac{1}{\beta^2}, \frac{1}{\gamma^2}$

if $OP^2 + OQ^2 + OR^2 = \frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} + \frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$
 $= 2(\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2})$
 $= 2K + K^2$ from A and B

c)



Join AB

$\angle PAB = \theta$ (exterior angle of cyclic quad)
 $\angle PCB = \theta$ (angles in same segment subtended by arc BP)

$\therefore DBCA$ is a cyclic quadrilateral
 (exterior $\angle$ $\angle B$ is interior opposite $\angle D$)

Q. 6 a)

i) $y = 4/3x, y = -4/3x$

ii) $\frac{2x}{9} - \frac{2y}{16} \frac{dy}{dx} = 0$

$$\frac{dy}{dx} = \frac{4 \sec \theta}{3 \tan \theta} \left(\frac{4}{3 \sin \theta} \right)$$

iii) $y = 4 \tan \theta = \frac{4 \sec \theta}{3 \tan \theta}$

$$3 \tan \theta y = 12 \tan^2 \theta = 4 \sec^2 \theta$$

$$4 \sec^2 \theta = 3 \tan^2 \theta + 12$$

$$x \cos \theta = 3 \sin \theta y + 12 \cos \theta$$

iv) Now $4x = -3y$

$$\therefore -3y = 3 \sin \theta y + 12 \cos \theta$$

$$\therefore y = \frac{-12 \cos \theta}{3 \sin \theta - 3}$$

$$\therefore x = \frac{3 \cos \theta}{\sin \theta + 1}$$

v) $\frac{3 \sec \theta}{2} = \frac{3 \cos \theta}{1 + \sin \theta}$

Simplifies to

$$6 \sin^2 \theta + 3 \sin \theta - 3 = 0$$

$$3(2 \sin^2 \theta - 1)(\sin \theta + 1)$$

$$\therefore \sin \theta = 1/2 \text{ or } -1$$

$$\sin \theta \neq -1 \text{ in } \frac{3 \cos \theta}{1 + \sin \theta}$$

$$\therefore \theta = \pi/6$$

b) $\int_0^{\pi/4} \tan \theta d\theta = \int_0^{\pi/4} \frac{\sin \theta}{\cos \theta} d\theta$

$$= -[\log_e(\cos \theta)]_0^{\pi/4}$$

$$= -[\log_e \frac{1}{\sqrt{2}}]$$

$$= \frac{1}{2} \log_e 2$$

$$ii) \int_0^{\pi/4} \tan^n \theta d\theta = \int_0^{\pi/4} \tan^{n-2} \theta \sec^2 \theta d\theta$$

$$= \int_0^{\pi/4} \tan^{n-2} \theta \sec^2 \theta d\theta$$

$$\therefore I_n + I_{n-2} = \left[\frac{\tan^{n-1} \theta}{n-1} \right]_0^{\pi/4}$$

$$= \frac{1}{n-1}$$

iii) $I_5 + I_3 = \frac{1}{4}$

$$I_3 + I_1 = \frac{1}{2}$$

$$\therefore I_5 - I_1 = -1/4$$

$$I_5 = -1/4 + I_1$$

$$\int_0^{\pi/4} \tan^5 \theta d\theta = \frac{1}{2} \log_e 2 - 1/4$$

$$\left[\tan^{-1}(3) - \tan^{-1}\left(\frac{1}{2}\right) \right]$$

$$\frac{3 - 1/2}{1 + 3 \cdot 1/2}$$

$$= 1$$

$$= \tan^{-1} 4$$

$$\tan^{-1}(3) - \tan^{-1}(1/2) = \pi/4$$

$$= 1 \text{ statement is true}$$

$$\text{Assume true for } n=k.$$

$$\tan^{-1} \frac{1}{2^k} = \tan^{-1}(2k+1) - \pi/4 \quad (A)$$

$$\sum_{r=1}^{k+1} \tan^{-1} \frac{1}{2^r} = \tan^{-1}(2k+3) - \pi/4$$

$$\sum_{r=1}^{k+1} \tan^{-1} \frac{1}{2^r} = \sum_{r=1}^k \tan^{-1} \frac{1}{2^r} + \tan^{-1} \frac{1}{2^{k+1}}$$

$$= \tan^{-1}(2k+1) - \pi/4 + \tan^{-1} \frac{1}{2^{k+1}}$$

$$\tan^{-1}(2k+3) - \pi/4 = \tan^{-1}(2k+1) - \pi/4 + \tan^{-1} \frac{1}{2^{k+1}}$$

$$\tan^{-1}(2k+3) = \tan^{-1}(2k+1) + \tan^{-1} \frac{1}{2^{k+1}}$$

$$\frac{2k+3 - (2k+1)}{1 + (2k+1)(2k+1)}$$

$$= \frac{2}{4k^2 + 8k + 4}$$

$$= \frac{1}{2(k+1)^2}$$

$$= \tan^{-1}(2k+3) - \tan^{-1}(2k+1)$$

$$= \tan^{-1} \frac{1}{2^{k+1}}$$

$$b) i) v = \frac{1}{2}(1-x^2)$$

$$\frac{dv}{dx} = -x$$

$$\text{Now } a = v \frac{dv}{dx} = \frac{x^3 - x}{2}$$

$$ii) \frac{1}{1+x} + \frac{1}{1-x} = \frac{1-x+1+x}{(1+x)(1-x)} = \frac{2}{1-x^2}$$

$$\frac{dx}{dt} = \frac{1-x^2}{2}$$

$$\int_0^x \frac{2}{1-x^2} dx = \int_0^t dt$$

$$\int_0^x \frac{1}{1+x} + \frac{1}{1-x} dx = t$$

$$\left[\log_e(1+x) - \log(1-x) \right]_0^x = t$$

$$\therefore t = \log_e \frac{1+x}{1-x}$$

$$e^t = \frac{1+x}{1-x}$$

$$e^t(1-x) = 1+x$$

$$x = \frac{e^t - 1}{e^t + 1} \left(\frac{1 - e^{-t}}{1 + e^t} \right)$$

iii) moves to right, slowing down. limiting position is $x=1$

$$c) i) x^3 + y^3 = 6xy$$

$$3x^2 + 3y^2 \frac{dy}{dx} = 6 \left[y \cdot 1 + x \frac{dy}{dx} \right]$$

$$\frac{dy}{dx} = \frac{2y - x^2}{y^2 - 2x}$$

$$ii) \frac{dy}{dx} = 0 \quad \therefore \frac{2y - x^2}{y^2 - 2x} = 0$$

$$y = \frac{x^2}{2}$$

$$\text{Subst (i)}$$

$$x^3 + \left(\frac{x^2}{2} \right)^3 = 6x \cdot \frac{x^2}{2}$$

$$16x^3 - 2x^3 = 0$$

$$x^3(16 - x^3) = 0$$

$$x = 0 \text{ or } \sqrt[3]{16}$$

$$\text{at } x=0 \frac{dy}{dx} \text{ is undefined}$$

$$\therefore x = \sqrt[3]{16}$$

$$a) i) S = \frac{1}{1+x}$$

$$ii) \int \frac{1}{1+x} dx = \int (1-x+x^2-x^3) dx$$

$$\log_e(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$\text{Let } x=1$$

$$\log_e 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

$$b) i) \int x \tan^{-1} x dx = \int \tan^{-1} x \frac{d(x^2/2)}{dx} dx$$

$$= \frac{x^2}{2} \tan^{-1} x - \int \frac{x^2}{2} \cdot \frac{1}{1+x^2} dx$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{x^2+1}{1+x^2} - \frac{1}{1+x^2} dx$$

$$= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \left[x - \tan^{-1} x \right]$$

$$= \frac{1}{2} [x^2+1] \tan^{-1} x - \frac{1}{2} x + c$$

$$iii) \int_0^1 x^n \tan^{-1} x dx = \int_0^1 x^{n-1} (x \tan^{-1} x) dx$$

$$= \int_0^1 x^{n-1} \frac{d}{dx} \left[\frac{1}{2} (x^2+1) \tan^{-1} x - \frac{1}{2} x \right] dx$$

$$= \left[x^{n-1} \left[\frac{1}{2} (x^2+1) \tan^{-1} x - \frac{1}{2} x \right] \right]_0^1 - \int_0^1 (n-1) x^{n-2} \left[\frac{1}{2} (x^2+1) \tan^{-1} x - \frac{1}{2} x \right] dx$$

$$I_n = \frac{\pi}{4} - \frac{1}{2} - \frac{n-1}{2} \int_0^1 x^n \tan^{-1} x + x^{n-2} \tan^{-1} x - x^{n-1} dx$$